

# Foundations of RL with Verifiable Rewards: Comparing Group Relative Policy Optimization with PPO and DPO

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## Abstract

GRPO, PPO, and DPO are all widely used methods for post-training language models, yet their theoretical relationships and relative merits are poorly understood. In this project we establish a formal comparison across three axes: the data format each method requires, the computational and statistical costs of the advantage or reward estimation procedure, and the conditions under which each achieves the best sample complexity for a given accuracy target. We show that all three methods optimize variants of the same KL-regularized reward maximization objective, but they differ in how they estimate the advantage signal: PPO uses a learned critic (low variance but subject to critic approximation bias), GRPO uses a group-relative empirical baseline (avoiding critic approximation bias; the population-normalized scalar surrogate has zero mean—a proved statement about the baseline estimator only—and the empirically normalized version has heuristic mean error  $O(1/G)$ ; separately, the group-mean baseline estimation error concentrates at rate  $O(1/\sqrt{G})$ ), and DPO implicitly estimates rewards via preference pairs (no online RL loop during optimization, but requires a specific paired-comparison data format). We characterize the bias–variance tradeoff between PPO and GRPO precisely, identify the regime in which each is preferable, and explain why DPO’s favorable sample complexity comes with fundamental limitations when rewards are verifiable and individual completions can be evaluated directly.

## 1 Introduction

GRPO is empirically successful but theoretically underexplored. Introduced by Shao et al. [2024] for training DeepSeekMath, GRPO has become the method of choice for training reasoning models including DeepSeek-R1 [DeepSeek-AI et al., 2025]. The key question is: *how does GRPO compare to PPO and DPO theoretically, and when should each be preferred?*

**Why the comparison matters.** All three methods, PPO [Schulman et al., 2017], DPO [Rafailov et al., 2023], and GRPO [Shao et al., 2024], are popular choices for aligning language model behavior, and practitioners routinely face the decision of which to use for a given task. The choice has significant practical consequences: PPO requires training a critic network (additional compute and potential instability); DPO requires preference pairs rather than individual outcome labels (a different annotation workflow); and GRPO requires multiple rollouts per prompt (higher inference cost but no critic). A principled theoretical comparison can guide this decision and explain the empirical performance differences observed in practice.

**The core distinction: data format and advantage source.** The three methods differ most fundamentally in how they obtain their training signal:

- **PPO** trains from individual (prompt, completion, reward) triples, using a learned critic  $V_\phi(s)$  to estimate the advantage  $A(s, a) = Q(s, a) - V(s)$  of each token. The critic reduces gradient variance but introduces bias when  $V_\phi$  misspecifies the true value function.
- **GRPO** also trains from individual reward evaluations, but avoids the critic by normalizing rewards within a group of  $G$  rollouts from the same prompt. The group mean replaces the critic as a prompt-specific baseline. This avoids critic approximation error but relies on a noisier finite-group estimate of the prompt baseline when  $G$  is small (the baseline estimation error is  $O(1/\sqrt{G})$  in the mean-square sense).
- **DPO** trains from *preference pairs*: for each prompt, a pair of completions  $(o^+, o^-)$  where  $o^+$  is preferred over  $o^-$  according to some human or oracle judgement. DPO eliminates the RL training loop entirely, fitting the policy to the preference data directly via a supervised cross-entropy loss. In its standard offline form it requires no reward model, no critic, and no online RL rollout loop during the optimization phase itself; however, obtaining or refreshing the preference-pair data may still require generating completions. It also requires the preference-pair data format, which is not directly available when rewards are verifiable scalars.

**Related work.** Schulman et al. [2017] introduced PPO and demonstrated its empirical effectiveness; Wang et al. [2020] analyzed the optimization landscape of policy gradient methods. Rafailov et al. [2023] showed that DPO optimizes a regularized reward maximization objective in closed form under the Bradley-Terry preference model, establishing that the optimal policy can be recovered without ever explicitly training a reward model. Azar et al. [2024] extended the theoretical analysis of DPO to handle more general preference models and derived sample complexity bounds.

Xiong et al. [2024] analyzed iterative preference learning, showing that on-policy variants (where the model generates its own preference pairs each iteration) can substantially outperform offline DPO due to improved distributional coverage. Kim et al. [2026] proved that on-policy preference learning converges exponentially faster than offline methods due to a coverage improvement mechanism, establishing a formal separation. Mroueh [2025] showed GRPO’s connection to KL-regularized contrastive learning, providing the unified view with DPO that we build on here. Mroueh et al. [2025] compared on-policy and off-policy GRPO, showing that both can be effective when properly configured. Tajwar et al. [2024] argued that preference fine-tuning should leverage on-policy data; their analysis of distributional alignment is directly relevant to understanding when GRPO (which is inherently on-policy) has an advantage over offline DPO.

## 2 Theoretical Framework

### 2.1 The Unified KL-Regularized Objective

All three methods, PPO, GRPO, and DPO, can be understood as optimizing variants of the same underlying objective, which we state carefully here.

**The common objective.** Let  $\pi(\cdot|s)$  be a policy over completions  $a \in \mathcal{A}$  given prompts  $s \in \mathcal{S}$ , and let  $r(s, a) \in \mathbb{R}$  be a reward function. The KL-regularized reward maximization objective is:

$$J_\beta(\pi) = \mathbb{E}_{s \sim \mu} [\mathbb{E}_{a \sim \pi(\cdot|s)} [r(s, a)]] - \beta \mathbb{E}_{s \sim \mu} [D_{\text{KL}}(\pi(\cdot|s) \parallel \pi_{\text{ref}}(\cdot|s))], \quad (1)$$

where  $\mu$  is the prompt distribution and  $\pi_{\text{ref}}$  is a reference policy. This objective has a unique closed-form maximizer:

$$\pi^*(a|s) = \frac{\pi_{\text{ref}}(a|s) \exp(r(s, a)/\beta)}{Z_\beta(s)}, \quad (2)$$

where  $Z_\beta(s) = \sum_{a'} \pi_{\text{ref}}(a'|s) \exp(r(s, a')/\beta)$ . All three methods target this same  $\pi^*$ ; they differ in how they estimate the gradient of  $J_\beta$  and update toward  $\pi^*$ .

**The three methods as estimators of the same target.** PPO and GRPO observe individual scalar rewards  $r(s, a)$  and compute gradient estimates; DPO observes *pairwise comparisons*  $o^+ \succ o^-$  and fits the log-ratio  $\log(\pi(o^+|s)/\pi_{\text{ref}}(o^+|s)) - \log(\pi(o^-|s)/\pi_{\text{ref}}(o^-|s))$  directly. The fundamental equivalence—that the DPO objective recovers the same  $\pi^*$  as reward-based methods under the Bradley-Terry model—justifies treating them as members of the same family.

## 2.2 Method-by-Method Analysis

### 2.2.1 PPO: Learned Critic with Low Variance

PPO estimates advantages using a learned critic (value function)  $V_\phi(s) \approx \mathbb{E}_\pi[R|s]$ . The advantage estimate for token  $a_t$  at state  $s_t$  is:

$$\hat{A}^{\text{PPO}}(s_t, a_t) = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (3)$$

or, using Generalized Advantage Estimation [Schulman et al., 2016], a  $\lambda$ -weighted mixture of  $k$ -step TD residuals.

**The critic’s role.** The critic  $V_\phi$  serves as a *state-dependent baseline* that reduces the variance of the policy gradient estimator from  $O(\sigma_R^2)$  (variance of the return) down to  $O(\sigma_A^2)$ , the variance of the residual advantage. For long-horizon tasks where  $\sigma_R^2$  is large, the variance reduction from a good critic can be enormous.

**The critic’s cost.** Training the critic introduces statistical cost (learning a function class of complexity  $d_V$ ), computational cost (extra forward and backward passes), and crucially a *bias* cost: if  $V_\phi$  misspecifies the true value function, the advantage estimate is biased by up to  $\varepsilon_V/(1-\gamma)$ , where  $\varepsilon_V = \sup_s |V_\phi(s) - V^\pi(s)|$  is the critic approximation error. When the horizon  $H$  is long ( $(1-\gamma)^{-1} \gg 1$ ) or the critic is imperfect ( $\varepsilon_V > 0$ ), this bias can be severe.

### 2.2.2 GRPO: Group-Relative Baseline, No Critic

GRPO replaces the learned critic with an empirical baseline computed from  $G$  i.i.d. rollouts of the same prompt. The group-normalized advantage is:

$$\hat{A}_i^{\text{GRPO}} = \frac{r_i - \bar{r}_G}{\sigma_{r,G}}, \quad \bar{r}_G = \frac{1}{G} \sum_{j=1}^G r_j, \quad \sigma_{r,G} = \sqrt{\frac{1}{G} \sum_{j=1}^G (r_j - \bar{r}_G)^2}. \quad (4)$$

The group mean  $\bar{r}_G$  estimates  $\mu^*(s) = \mathbb{E}_{\pi(\cdot|s)}[r(s, \cdot)]$ , which is exactly the quantity PPO’s critic tries to learn (in the episode-level setting with  $\gamma = 1$ ). The key difference is that GRPO *estimates*  $\mu^*(s)$  fresh from  $G$  rollouts each time, with no function-approximation error.

### Variance of the GRPO advantage.

**Theorem 2.1** (Variance of the GRPO Advantage Estimator). *Let  $\sigma^{*2}(s) = \text{Var}_{\pi(\cdot|s)}[r(s, \cdot)]$  be the population reward variance for prompt  $s$ .*

(i) Numerator variance (exact, all  $G \geq 2$ ).

$$\text{Var}[r_i - \bar{r}_G] = \sigma^{*2}(s) \left(1 - \frac{1}{G}\right), \quad (5)$$

*showing that the group baseline removes exactly a fraction  $1/G$  of the reward variance for every finite  $G \geq 2$ .*

(ii) Normalized advantage variance (population normalization, exact). *For the surrogate with denominator fixed to  $\sigma^*(s)$ :*

$$\text{Var}[\hat{A}_i^{\text{GRPO}}] = 1 - \frac{1}{G}. \quad (6)$$

*Remark 2.2* (Empirical normalization: asymptotic approximation). For the empirically normalized estimator (random denominator  $\sigma_{r,G}$ ), a delta-method expansion suggests heuristically that  $\text{Var}[\hat{A}_i^{\text{GRPO}}] = 1 + O(1/G)$  for large  $G$ . This is an asymptotic approximation, not a proved finite- $G$  bound. Both the exact expression  $1 - 1/G$  (population) and the approximation  $1 + O(1/G)$  (empirical) approach 1 from opposite sides as  $G \rightarrow \infty$ .

**Intuition.** Equation (5) shows that increasing  $G$  does *not* decrease the variance of the advantage numerator—it increases it toward  $\sigma^{*2}$ . What larger  $G$  provides instead is a more *accurate* advantage estimate: the group mean  $\bar{r}_G$  concentrates around  $\mu^*$  at rate  $O(1/\sqrt{G})$  (baseline estimation error), so  $\hat{A}_i$  converges to the true normalized advantage. In practice, most of the estimation-accuracy benefit is captured by  $G \in [4, 16]$ .

### 2.2.3 DPO: Implicit Reward via Preference Pairs

DPO observes *preference pairs*  $(o^+, o^-)$  for each prompt  $s$ . Under the Bradley-Terry preference model [Rafailov et al., 2023]:

$$P(o^+ \succ o^- | s) = \sigma(r^*(s, o^+) - r^*(s, o^-)), \quad (7)$$

where  $r^*$  is the true underlying reward. Substituting the optimal-policy reparametrization  $r^*(s, a) = \beta \log(\pi^*(a|s)/\pi_{\text{ref}}(a|s)) + \beta \log Z_\beta(s)$  into the Bradley-Terry likelihood (where the  $\log Z_\beta$  terms cancel in the difference), the DPO training objective is:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta) = -\mathbb{E}_{(s, o^+, o^-)} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(o^+|s)}{\pi_{\text{ref}}(o^+|s)} - \beta \log \frac{\pi_\theta(o^-|s)}{\pi_{\text{ref}}(o^-|s)} \right) \right], \quad (8)$$

a standard binary cross-entropy loss requiring no online RL loop during optimization.

**Why DPO does not require an online RL loop.** The DPO loss is a supervised learning objective over a fixed dataset of preference pairs. Minimizing it does not require online rollout generation, a reward model, or a separate critic *during optimization*. The price is the data format requirement: DPO needs *paired* comparisons, not individual scalar rewards. For verifiable RLVR tasks where completions are independently labeled correct/incorrect, preference pairs must be constructed synthetically (e.g., pairing each correct completion with a randomly chosen incorrect one), which introduces additional design choices.

**DPO is inherently offline.** The basic DPO formulation is *offline*: preference pairs are drawn from a fixed dataset and the policy never generates new data. As shown by Xiong et al. [2024] and Kim et al. [2026], offline methods can suffer from distributional mismatch. On-policy variants (iterative DPO) mitigate this at the cost of requiring rollouts, bringing the computational cost closer to GRPO.

### 2.3 Formal Comparison: Sample Complexity

**Proposition 2.3** (Comparative Scaling of Sample Complexity). (Informal.) *Let  $\varepsilon > 0$  be the target gradient-norm accuracy. Define  $C_\pi$  as a combined policy-optimization constant (absorbing smoothness, initialization gap, and score-function bound);  $d_V$  as the statistical complexity of the critic function class;  $G$  as the GRPO group size; and  $\varepsilon_V$  as the best-achievable critic approximation error. Under standard smoothness and realizability assumptions, the rollout counts to reach  $\varepsilon$ -accuracy scale as:*

$$n_{\text{PPO}} = O\left(\frac{d_V + C_\pi}{\varepsilon^2}\right) + \underbrace{O\left(\frac{\varepsilon_V}{(1-\gamma)\varepsilon}\right)}_{\text{irreducible bias floor}}, \quad (9)$$

$$n_{\text{GRPO}} = O\left(\frac{C_\pi}{\varepsilon^2}\right) \quad (\text{independent of } G), \quad (10)$$

$$n_{\text{DPO}} = O\left(\frac{d_\pi}{\varepsilon^2}\right) \quad (\text{offline, assuming adequate coverage}). \quad (11)$$

**Interpreting the bounds.** For PPO (9), the additive bias term captures the irreducible accuracy floor: if the critic class cannot represent  $V^\pi$ , PPO cannot converge below  $O(\varepsilon_V/(1-\gamma))$  regardless of sample size. For GRPO (10), the total rollout cost is independent of group size  $G$  because the  $G$ -fold variance reduction from larger groups exactly cancels the  $G$ -fold rollout cost per gradient step. For DPO (11), the favorable rate counts *preference pairs*, each requiring two completions; the true completion count is  $2d_\pi/\varepsilon^2$ , comparable to GRPO with  $G = 2$ . This rate also assumes adequate offline coverage, which frequently fails in practice.

*Remark 2.4* (When each method is preferable). **(1) PPO is preferable** when the critic can be trained accurately ( $\varepsilon_V \approx 0$ ) and the critic class is simple ( $d_V \ll d_\pi$ ). This is most likely when rewards are dense and the state structure is simple.

**(2) GRPO is preferable** when the critic is hard to train (large  $\varepsilon_V$ ) or dense step-level rewards are unavailable (sparse rewards, as in RLVR). For verifiable tasks with binary sparse rewards, the irreducible critic bias  $\varepsilon_V/(1-\gamma)$  of PPO becomes the bottleneck, while GRPO’s baseline bias vanishes with more rollouts.

**(3) DPO is preferable** when paired preference data is available and the training distribution is close to the deployment distribution. For verifiable reward tasks, DPO’s advantage diminishes because preference pairs must be constructed from scalar rewards, and the offline limitation becomes significant.

### 2.4 The Bias–Variance Tradeoff Between PPO and GRPO

**Theorem 2.5** (Bias–Variance Profile: PPO vs. GRPO). *For the episode-level RLVR setting ( $\gamma = 1$ , binary rewards  $r \in \{0, 1\}$ ):*

GRPO advantage (group size  $G$ , prompt  $s$ , completion  $o_i$ ). Here “Bias” means  $\mathbb{E}[\hat{A}_i^{\text{GRPO}}]$ : the mean of the estimator, which we require to equal zero for unbiased baseline subtraction (not unbiasedness with respect to the true  $Q$ -function advantage  $A^*(s, a)$ , which is a distinct and stronger condition).

$$\mathbb{E}[\hat{A}_i^{\text{GRPO}}] = \begin{cases} 0 & (\text{population normalization: proved exactly}), \\ O(1/G) & (\text{empirical normalization: heuristic}), \end{cases} \quad (12)$$

$$\text{Var}[\hat{A}_i^{\text{GRPO}}] = \begin{cases} 1 - 1/G & (\text{population normalization: proved exactly}), \\ 1 + O(1/G) & (\text{empirical normalization: heuristic approximation}). \end{cases} \quad (13)$$

PPO advantage (critic approximation error  $\varepsilon_V = \sup_s |V_\phi(s) - V^\pi(s)|$ ).

$$|\mathbb{E}[\hat{A}^{\text{PPO}}(s, a)] - A^*(s, a)| \leq \frac{\varepsilon_V}{1 - \gamma}, \quad (14)$$

$$\text{Var}[\hat{A}^{\text{PPO}}(s, a)] = O(\sigma_A^{*2}) \leq O(\sigma_R^2), \quad (15)$$

where  $A^*(s, a) = Q^\pi(s, a) - V^\pi(s)$  is the true advantage,  $\sigma_A^{*2}$  is its variance, and  $\sigma_R^2$  is the variance of the return.

**When bias dominates: the case for GRPO.** The PPO advantage bias  $\varepsilon_V/(1 - \gamma)$  grows without bound as  $\gamma \rightarrow 1$  or as  $\varepsilon_V$  grows. For RLVR with long reasoning chains and binary sparse rewards, both conditions are severe. GRPO, which avoids critic approximation bias entirely, is clearly preferable in this regime.

**When variance dominates: the case for PPO.** The PPO advantage variance  $O(\sigma_A^{*2})$  can be much smaller than  $O(1)$  (the GRPO variance) when the value function captures most of the variation in returns. In dense-reward MDPs, a good critic reduces variance by orders of magnitude, and  $\varepsilon_V \approx 0$ , so PPO’s bias term vanishes and it converges faster than GRPO.

*Remark 2.6* (The group size as an estimation-accuracy knob). From Theorem 2.1, the GRPO advantage numerator variance  $\sigma^{*2}(1 - 1/G)$  increases toward  $\sigma^{*2}$  as  $G$  grows. What larger  $G$  provides is a more accurate advantage estimate:  $\bar{r}_G$  concentrates around  $\mu^*$  at rate  $O(1/\sqrt{G})$ . The analogy with PPO: larger  $G$  plays the same role as a better critic, reducing advantage *estimation error*, but without training cost and without an irreducible misspecification floor.

## 2.5 DPO and GRPO: Unified View and Key Differences

Mroueh [2025] established that GRPO with verifiable rewards can be written as a KL-regularized contrastive loss, structurally similar to DPO. For a group of  $G = 2$  completions ( $o^+$  with  $r = 1$ ,  $o^-$  with  $r = 0$ ), the GRPO group-normalized advantages reduce to  $\hat{A}^+ = +1$  and  $\hat{A}^- = -1$ , so the GRPO policy gradient is proportional to:

$$\nabla_\theta [\log \pi_\theta(o^+|s) - \log \pi_\theta(o^-|s)], \quad (16)$$

the gradient of the log-odds ratio between the correct and incorrect completion. The DPO gradient of (8) is proportional to  $\nabla_\theta [\log \pi_\theta(o^+|s)/\pi_{\text{ref}}(o^+|s) - \log \pi_\theta(o^-|s)/\pi_{\text{ref}}(o^-|s)]$ , which points in the same direction. When  $\pi_\theta \approx \pi_{\text{ref}}$ , the log-ratio terms vanish and the GRPO and DPO gradients are approximately proportional.



**Key differences despite structural similarity.**

- (1) **Data source.** GRPO generates its own comparison pairs through online rollouts; DPO uses a fixed offline dataset. GRPO is therefore inherently on-policy, avoiding the distributional mismatch identified as the key weakness of offline DPO.
- (2) **Reference policy treatment.** DPO explicitly computes  $\log(\pi_\theta(o)/\pi_{\text{ref}}(o))$  for both  $o^+$  and  $o^-$ , requiring two forward passes through  $\pi_{\text{ref}}$  per training step. GRPO subtracts only the group mean  $\bar{r}_G$  (a scalar), making GRPO computationally cheaper per step when  $G \leq 4$ .
- (3) **Signal richness.** With  $G > 2$ , GRPO uses more than one positive–negative pair per prompt, computing a richer within-group advantage signal that single-pair DPO cannot.

| Property                      | PPO                           | GRPO                       | DPO                                          |
|-------------------------------|-------------------------------|----------------------------|----------------------------------------------|
| Data format                   | Scalar reward                 | Scalar reward              | Preference pairs                             |
| Critic required               | Yes ( $d_V$ cost)             | No                         | No                                           |
| On-policy                     | Yes                           | Yes                        | No (offline)                                 |
| Advantage bias                | $O(\varepsilon_V/(1-\gamma))$ | 0 (pop.) / $O(1/G)$ (emp.) | N/A                                          |
| Advantage variance (unnorm.)  | $O(\sigma_A^{*2})$            | $\sigma^{*2}(1-1/G)$       | N/A                                          |
| Extra compute per prompt      | $2\times$ (critic)            | $G\times$ (rollouts)       | $2\times$ ( $\pi_{\text{ref}}$ forward pass) |
| Works with verifiable rewards | Yes                           | Yes                        | With pair construction                       |
| Preferred regime              | Dense reward, good critic     | Sparse/binary reward       | Preference data available                    |

Table 1: Comparison of PPO, GRPO, and DPO across key theoretical and practical dimensions.  $\varepsilon_V$ : critic approximation error;  $\sigma_A^{*2}$ : true advantage variance;  $\sigma^{*2}$ : population reward variance;  $G$ : group size. GRPO bias is 0 under population normalization (denominator  $\sigma^*(s)$ ) and  $O(1/G)$  under empirical normalization (denominator  $\sigma_{r,G}$ ). GRPO advantage variance is shown unnormalized for comparability with PPO.

### 3 Experimental Design

#### 3.1 E2.7: Head-to-Head Comparison Under Matched Compute

**Setup.** Implement all three methods on GSM8K [Cobbe et al., 2021] and HumanEval [Chen et al., 2021] using Llama-3 8B as the base model. For DPO on verifiable tasks, construct preference pairs by pairing each correct completion with a randomly sampled incorrect completion from the same batch. Match total completion count across methods: PPO uses 1 rollout per prompt (plus critic); GRPO uses  $G = 8$  rollouts per prompt; DPO uses 2 rollouts per prompt (one positive, one negative).

**Measurements.** For each method: (i) final test accuracy at a fixed total rollout budget; (ii) training stability (variance of reward across iterations); (iii) convergence speed (accuracy vs. rollout count); (iv) the observed advantage estimation error  $|\hat{A} - A^{\text{MC}}|$  where  $A^{\text{MC}}$  is estimated by 1000 Monte Carlo rollouts.

**Theoretical predictions.** Remark 2.4 predicts GRPO should outperform PPO on the binary-reward RLVR tasks (where PPO’s critic is hard to train) and outperform offline DPO (due to on-policy coverage). The baseline estimation error plot should show PPO with non-zero irreducible critic-approximation bias and GRPO’s group-mean error decaying as  $O(1/\sqrt{G})$ .

### 3.2 E2.8: Critic Quality and the PPO–GRPO Crossover

**Setup.** Train PPO with varying critic capacity: (a) no critic (REINFORCE), (b) small critic (2-layer MLP), (c) medium critic (same architecture as policy), (d) large critic ( $2\times$  policy size). For each, measure critic approximation error  $\varepsilon_V$  on a held-out value estimation task and compare final policy performance against GRPO with  $G \in \{4, 8, 16\}$ .

**Measurements.** For each (critic size,  $G$ ) pair: (i) final accuracy; (ii) estimated critic error  $\varepsilon_V$ ; (iii) estimated advantage bias for PPO (compared with Monte Carlo ground truth); (iv) sample efficiency curves.

**Theoretical prediction.** Theorem 2.5 predicts a crossover: for small critic error  $\varepsilon_V < \varepsilon_V^*$  (accurate critic), PPO should outperform GRPO with small  $G$ ; for large  $\varepsilon_V > \varepsilon_V^*$ , GRPO is preferable. The crossover value is approximately

$$\varepsilon_V^* \approx (1 - \gamma) \sqrt{\sigma^{*2}(1 - 1/G) - \sigma_A^{*2}}.$$

### 3.3 E2.9: Verifiable Rewards vs. Preference Data

**Setup.** Compare DPO (trained on constructed preference pairs from RLVR outcomes) with GRPO (trained directly on verifiable rewards) under three data regimes: (a) full labels (all completions labeled), (b) sparse labels (10% labeled), (c) noisy labels (10% of correct labels flipped to incorrect).

**Measurements.** For each condition: (i) final accuracy; (ii) calibration of the implicit reward  $\hat{r}(s, a) = \beta \log(\pi_\theta(a|s)/\pi_{\text{ref}}(a|s))$  (correlation with true correctness); (iii) distributional coverage (how closely the training completion distribution matches the evaluation distribution).

**Theoretical prediction.** GRPO should be more robust to label noise (group normalization is self-contained within each batch) and to sparse labels (any correct/incorrect label directly provides a gradient signal). DPO’s implicit reward calibration should be sensitive to preference pair quality: noisy pairs degrade the  $\log(\pi/\pi_{\text{ref}})$  alignment with the true reward, eroding the DPO objective’s connection to  $\pi^*$ .

## 4 Tasks

1. **Implement all three methods.** Implement PPO (with critic), GRPO, and DPO using a common Llama-3 8B backbone. For DPO on verifiable tasks, implement the synthetic pair construction procedure.
2. **Run E2.7: head-to-head comparison.** Run all three methods on GSM8K and HumanEval under matched compute budgets. Report all measurements (accuracy, training stability, convergence speed, advantage estimation error) with standard deviations over 3 seeds.
3. **Run E2.8: critic quality sweep.** Implement PPO with varying critic architectures (a)–(d) and run the full sweep against GRPO with  $G \in \{4, 8, 16\}$ . Produce the crossover plot of critic error  $\varepsilon_V$  vs. accuracy and verify the predicted crossover value  $\varepsilon_V^*$ .



4. **Run E2.9: label regime comparison.** Run GRPO and DPO under the three label regimes (full, sparse, noisy) and report accuracy, reward calibration, and distributional coverage metrics.

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